



## Project

Database Fundamental 1 (CS2231)

## Project Name

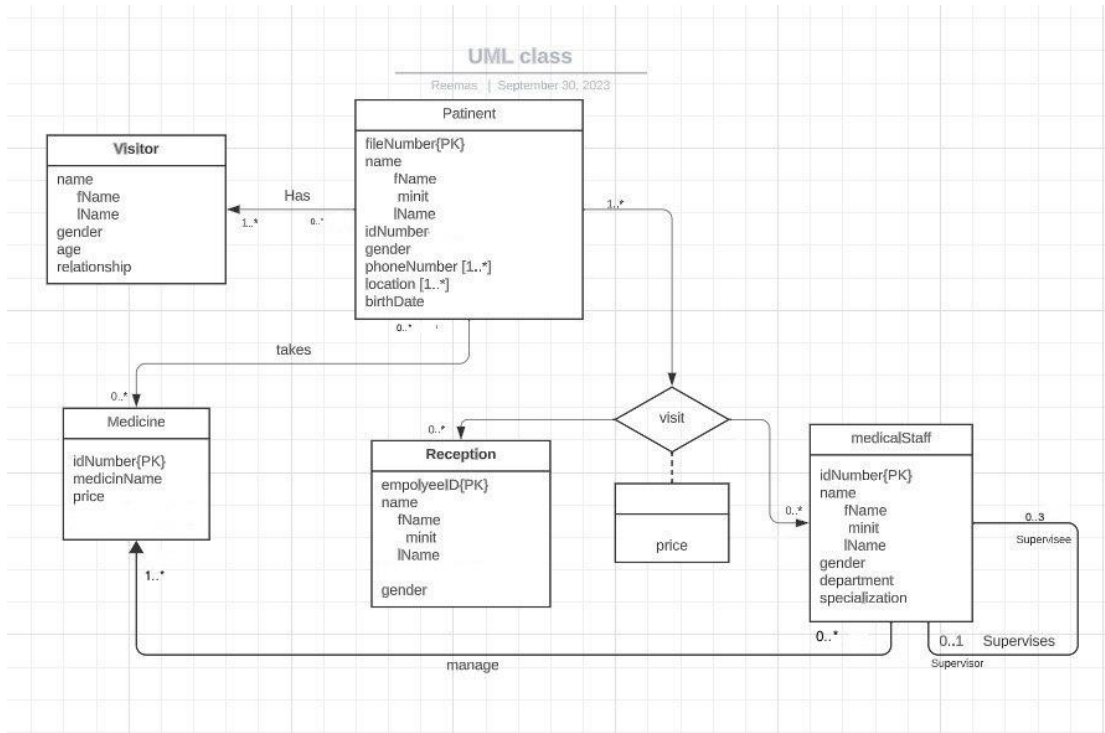
Hospital Database

| members of the group |            |                                               |
|----------------------|------------|-----------------------------------------------|
| Student name         | Student ID | Task                                          |
| أثير الحارثي         | 444006152  | Description + Final Mapping + SQL script      |
| ريماس ذبيان          | 444006605  | UML notation Step by step Mapping+ SQL script |
| غادة الحارثي         | 444006163  | Chen's notation +Normalization + SQL script   |

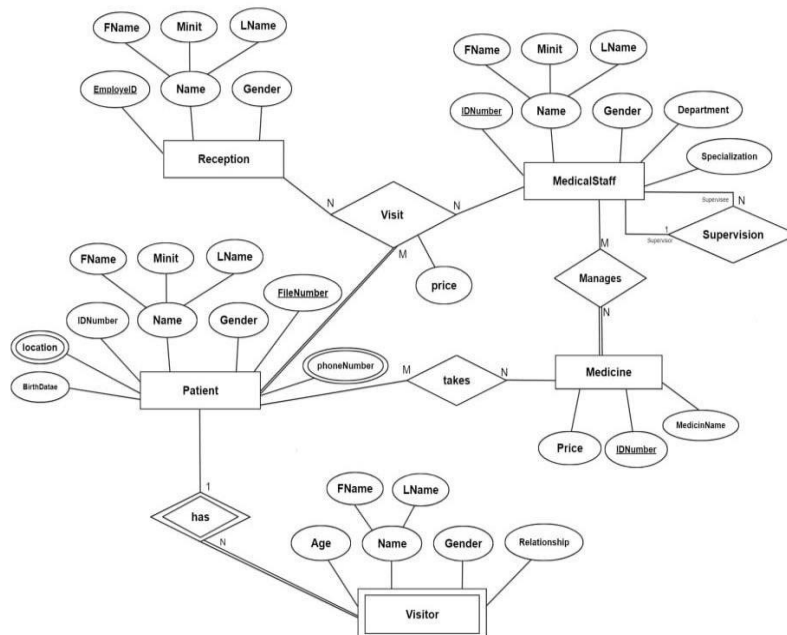
### Description:

- In the hospital, each patient visit medical staff and reception.
- Each visit has price.
- Each patient may has a family companion or visitor.
- The medicine is given to the patient but not all patients take medicine.
- When a patient needs medication, it is managed by medical staff.
- each medical staff has supervisor from medical staff. • Each medical staff has their first name, minit, last name gender, department, unique ID number, and specialty .
- Each patient has first name, minit, last name, ID number, gender, unique file number, several phone number, several location, and date of birth.
- The reception contains the first name, minit, last name, and gender, unique employee ID , file number
- Every visitor has first name, last name, gender, age, relationship. • Each medicine has a medicine name, unique ID number and price and patient name.

## UML Notation



## Chen's Notation



## ER- to-Relational Mapping:

### STEP 1:

#### Mapping of Regular / Strong Entity

##### Patient

|                   |          |       |       |       |        |          |
|-------------------|----------|-------|-------|-------|--------|----------|
| <u>fileNumber</u> | birhDate | fName | minit | lName | gender | idNumber |
|-------------------|----------|-------|-------|-------|--------|----------|

##### Reception

|                   |       |       |       |        |
|-------------------|-------|-------|-------|--------|
| <u>employeeID</u> | fName | minit | lName | gender |
|-------------------|-------|-------|-------|--------|

##### MedicalStaff

|                 |       |       |       |        |            |                |
|-----------------|-------|-------|-------|--------|------------|----------------|
| <u>idNumber</u> | fName | minit | lName | gender | department | specialization |
|-----------------|-------|-------|-------|--------|------------|----------------|

##### Medicine

|                 |              |       |
|-----------------|--------------|-------|
| <u>idNumber</u> | medicineName | price |
|-----------------|--------------|-------|

### STEP 2:

#### Mapping of Weak Entity:

##### Visitor

|                    |              |       |        |     |              |
|--------------------|--------------|-------|--------|-----|--------------|
| <u>pFileNumber</u> | <u>fName</u> | lName | gender | age | relationship |
|--------------------|--------------|-------|--------|-----|--------------|

### STEP 3:

#### Mapping of 1:1 relationship types

None

#### Step 4: Mapping of 1:N relationship types

##### SUPERVISION relationship

###### MedicalStaff

|                 |       |       |       |        |            |                |         |
|-----------------|-------|-------|-------|--------|------------|----------------|---------|
| <u>idNumber</u> | fName | minit | lName | gender | department | specialization | superID |
|-----------------|-------|-------|-------|--------|------------|----------------|---------|

#### Step 5: Mapping of N:M relationship types

##### TAKES relationship

###### Patient

|                   |          |       |       |       |        |          |
|-------------------|----------|-------|-------|-------|--------|----------|
| <u>fileNumber</u> | birhDate | fName | minit | lName | gender | idNumber |
|-------------------|----------|-------|-------|-------|--------|----------|

###### Medicine

|                 |              |       |
|-----------------|--------------|-------|
| <u>idNumber</u> | medicineName | price |
|-----------------|--------------|-------|

###### TAKES

|                    |                   |
|--------------------|-------------------|
| <u>pFileNumber</u> | <u>medicineID</u> |
|--------------------|-------------------|

##### MANAGES relationship type

###### MedicalStaff

|                 |       |       |       |        |            |                |
|-----------------|-------|-------|-------|--------|------------|----------------|
| <u>idNumber</u> | fName | minit | lName | gender | department | specialization |
|-----------------|-------|-------|-------|--------|------------|----------------|

###### Medicine

|                 |              |       |
|-----------------|--------------|-------|
| <u>idNumber</u> | medicineName | price |
|-----------------|--------------|-------|

MANAGES

|                  |                   |
|------------------|-------------------|
| <u>medicalID</u> | <u>medicineID</u> |
|------------------|-------------------|

## Step 6: Mapping of Multivalued Attributes

### PatientLoc Multivalued Attributes

#### Patient

|                   |          |       |       |       |        |          |
|-------------------|----------|-------|-------|-------|--------|----------|
| <u>fileNumber</u> | birhDate | fName | minit | lName | gender | idNumber |
|-------------------|----------|-------|-------|-------|--------|----------|

|                      |                 |
|----------------------|-----------------|
| <u>pFileNumberID</u> | <u>location</u> |
|----------------------|-----------------|

PatientLoc

### PatientPhone Multivalued Attributes

#### Patient

|                   |          |       |       |       |        |          |
|-------------------|----------|-------|-------|-------|--------|----------|
| <u>fileNumber</u> | birhDate | fName | minit | lName | gender | idNumber |
|-------------------|----------|-------|-------|-------|--------|----------|

|                      |                    |
|----------------------|--------------------|
| <u>pFileNumberID</u> | <u>phoneNumber</u> |
|----------------------|--------------------|

PatientPhone

## Step 7: Mapping of N-ary Relationship

### VISIT relationship type

#### Patient

|                   |          |       |       |       |        |          |
|-------------------|----------|-------|-------|-------|--------|----------|
| <u>fileNumber</u> | birhDate | fName | minit | lName | gender | idNumber |
|-------------------|----------|-------|-------|-------|--------|----------|

#### Reception

|                   |       |       |       |        |
|-------------------|-------|-------|-------|--------|
| <u>employeeID</u> | fName | minit | lName | gender |
|-------------------|-------|-------|-------|--------|

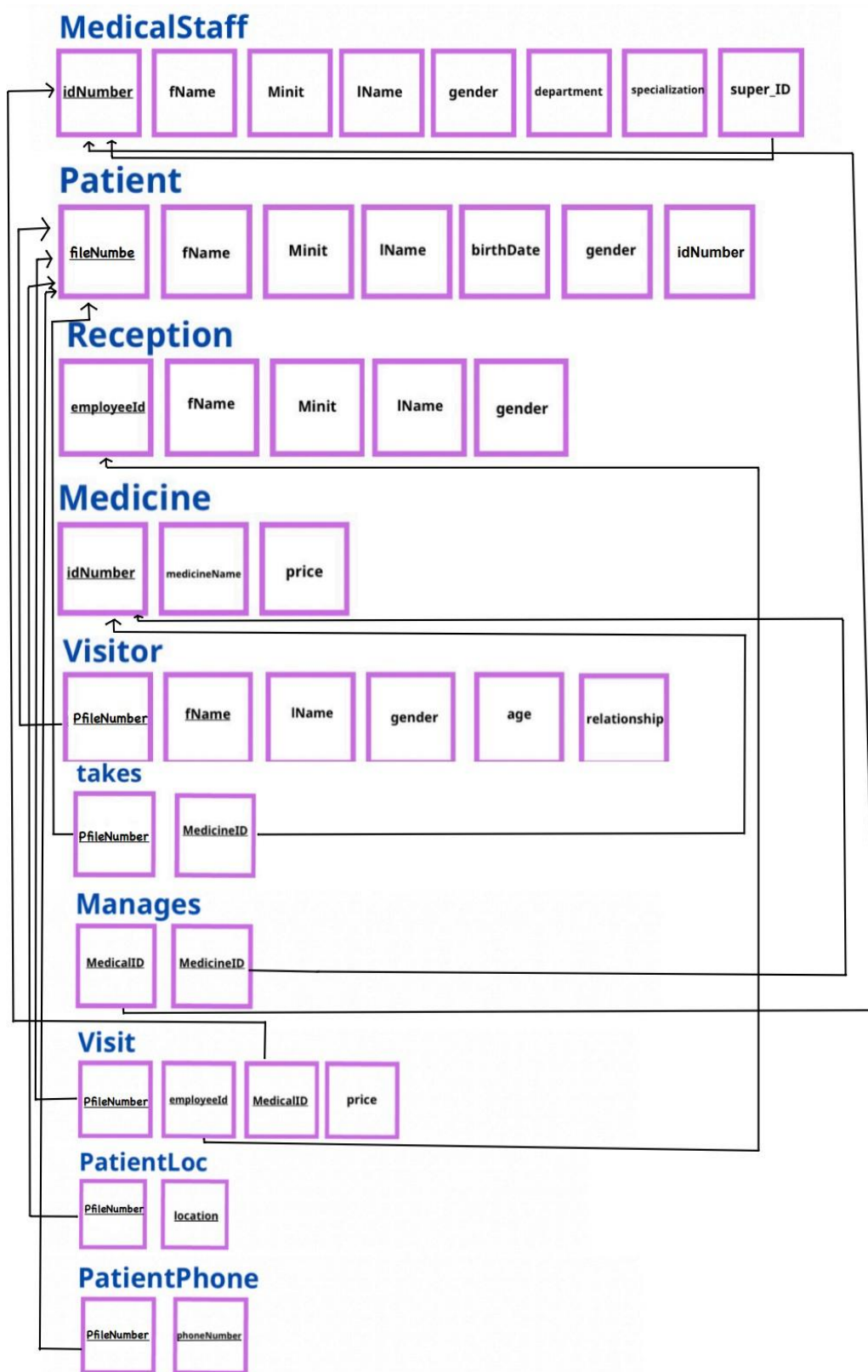
#### MedicalStaff

|                 |       |       |       |        |            |                |
|-----------------|-------|-------|-------|--------|------------|----------------|
| <u>idNumber</u> | fName | minit | lName | gender | department | specialization |
|-----------------|-------|-------|-------|--------|------------|----------------|

#### Visit

|                   |                   |                 |       |
|-------------------|-------------------|-----------------|-------|
| <u>fileNumber</u> | <u>employeeID</u> | <u>idNumber</u> | price |
|-------------------|-------------------|-----------------|-------|

## Final Mapping:



# Database Normalization:

All the Table are in:

- 1NF (No multivalued attributes),
- 2NF (No partial dependency),
- 3NF (No transitive dependency)

## MedicalStaff

|                 |       |       |       |        |            |                |         |
|-----------------|-------|-------|-------|--------|------------|----------------|---------|
| <u>idNumber</u> | fName | minit | lName | gender | department | specialization | superID |
|-----------------|-------|-------|-------|--------|------------|----------------|---------|

## Patient

|                   |          |       |       |       |        |          |
|-------------------|----------|-------|-------|-------|--------|----------|
| <u>fileNumber</u> | birhDate | fName | minit | lName | gender | idNumber |
|-------------------|----------|-------|-------|-------|--------|----------|

## Reception

|                   |       |       |       |        |
|-------------------|-------|-------|-------|--------|
| <u>employeeID</u> | fName | minit | lName | gender |
|-------------------|-------|-------|-------|--------|

## Medicine

|                 |              |       |
|-----------------|--------------|-------|
| <u>idNumber</u> | medicineName | price |
|-----------------|--------------|-------|

## Visitor

|                    |              |       |        |     |              |
|--------------------|--------------|-------|--------|-----|--------------|
| <u>pFileNumber</u> | <u>fName</u> | lName | gender | age | relationship |
|--------------------|--------------|-------|--------|-----|--------------|

## Takes

|                    |                   |
|--------------------|-------------------|
| <u>pFileNumber</u> | <u>medicineID</u> |
|--------------------|-------------------|

## Manages

|                  |                   |
|------------------|-------------------|
| <u>medicalID</u> | <u>medicineID</u> |
|------------------|-------------------|

## Visit

|                   |                   |                 |       |
|-------------------|-------------------|-----------------|-------|
| <u>fileNumber</u> | <u>employeeID</u> | <u>idNumber</u> | price |
|-------------------|-------------------|-----------------|-------|

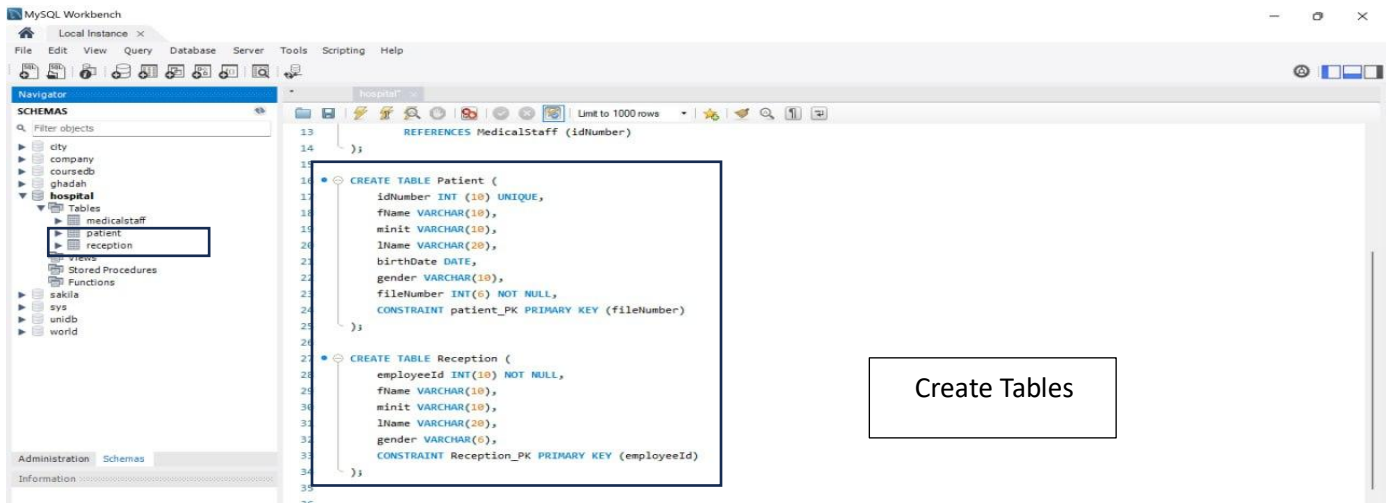
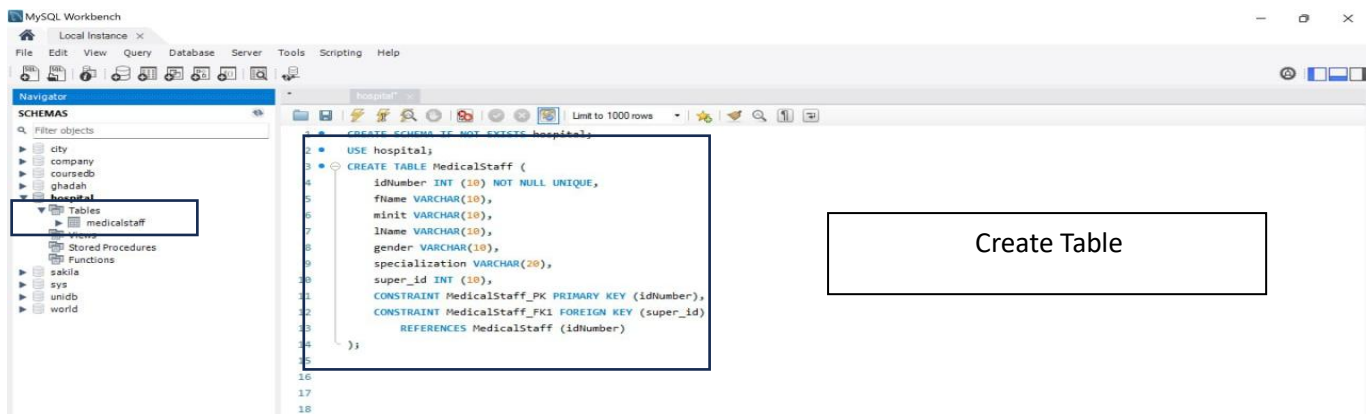
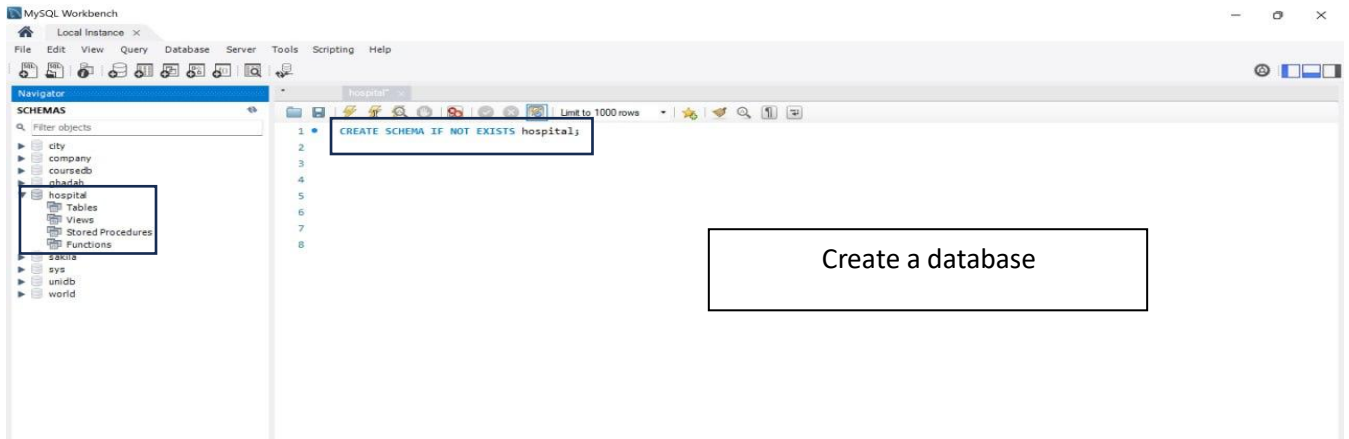
## PatientLoc

|                    |                 |
|--------------------|-----------------|
| <u>pFileNumber</u> | <u>location</u> |
|--------------------|-----------------|

## PatientPhone

|                    |                    |
|--------------------|--------------------|
| <u>pFileNumber</u> | <u>phoneNumber</u> |
|--------------------|--------------------|

# SQL command



MySQL Workbench

Local Instance x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- city
- company
- coursedb
- ghadah
- hospital
  - Tables
    - medicalstaff
    - medicine
    - patient
    - reception
    - visitor
  - Views
  - Stored Procedures
  - Functions
  - sakila
  - sys
  - unbdb
  - world

Administration Schemas Information

No object selected

hospital

Limit to 1000 rows

```

28 employeeId INT(10) NOT NULL,
29 fName VARCHAR(10),
30 mInit VARCHAR(10),
31 lName VARCHAR(20),
32 gender VARCHAR(6),
33 CONSTRAINT Reception_PK PRIMARY KEY (employeeId)
34 );
35
36 CREATE TABLE Medicine(
37 idNumber INT(10) NOT NULL ,
38 medicineName VARCHAR(20),
39 price INT(5) ,
40 CONSTRAINT Medicine_PK PRIMARY KEY (idNumber)
41 );
42
43 CREATE TABLE Visitor(
44 pFileNumber INT (10) NOT NULL ,
45 fName VARCHAR (10) NOT NULL ,
46 lName VARCHAR(20),
47 gender VARCHAR (10) ,
48 age INT (2) CHECK (age > 18),
49 relationship VARCHAR (10),
50 CONSTRAINT Visitor_PK PRIMARY KEY (pFileNumber, fName),
51 CONSTRAINT Visitor_FK1 FOREIGN KEY (pFileNumber) REFERENCES Patient(fileNumber)
52 );

```

Create Tables

MySQL Workbench

Local Instance x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- city
- company
- coursedb
- ghadah
- hospital
  - Tables
    - manages
    - medicalstaff
    - medicine
    - patient
    - reception
    - takes
    - visitor
  - Views
  - Stored Procedures
  - Functions
  - sakila
  - sys
  - unbdb
  - world

Administration Schemas Information

No object selected

hospital

Limit to 1000 rows

```

50 CONSTRAINT Visitor_PK PRIMARY KEY (pFileNumber, fName),
51 CONSTRAINT Visitor_FK1 FOREIGN KEY (pFileNumber) REFERENCES Patient(fileNumber)
52 );
53
54 CREATE TABLE Takes(
55 pFileNumber INT (6),
56 MedicineID INT (10),
57 CONSTRAINT Takes_PK PRIMARY KEY (pFileNumber, MedicineID),
58 CONSTRAINT Takes_FK1 FOREIGN KEY (pFileNumber) REFERENCES Patient(fileNumber),
59 CONSTRAINT Takes_FK2 FOREIGN KEY (MedicineID) REFERENCES Medicine(idNumber)
60 );
61
62 CREATE TABLE Manages(
63 medicalID INT (10) NOT NULL,
64 MedicineID INT (10) NOT NULL,
65 CONSTRAINT Manages_PK PRIMARY KEY (medicalID, MedicineID),
66 CONSTRAINT Manages_FK1 FOREIGN KEY (medicalID) REFERENCES MedicalStaff(idNumber),
67 CONSTRAINT Manages_FK2 FOREIGN KEY (MedicineID) REFERENCES Medicine(idNumber)
68 );
69

```

Create Tables

MySQL Workbench

Local Instance x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- city
- company
- coursedb
- ghadah
- hospital
  - Tables
    - manages
    - medicalstaff
    - medicine
    - patient
    - patientloc
    - patientphone
    - reception
    - takes
    - visit
    - visitor
  - Views
  - Stored Procedures
  - Functions
  - sakila
  - sys
  - unbdb
  - world

Administration Schemas Information

No object selected

hospital

Limit to 1000 rows

```

69
70 CREATE TABLE Visit (
71 pFileNumber INT(6) NOT NULL,
72 employeeId INT(10) NOT NULL,
73 medicalID INT(10) NOT NULL,
74 price INT(8),
75 CONSTRAINT Visit_PK PRIMARY KEY (pFileNumber , employeeId , medicalID),
76 CONSTRAINT Visit_FK1 FOREIGN KEY (pFileNumber) REFERENCES Patient (fileNumber),
77 CONSTRAINT Visit_FK2 FOREIGN KEY (medicalID) REFERENCES MedicalStaff (idNumber),
78 CONSTRAINT Visit_FK3 FOREIGN KEY (employeeId) REFERENCES Reception (employeeId)
79 );
80
81 CREATE TABLE PatientLoc(
82 pFileNumber INT (6) NOT NULL,
83 location VARCHAR (40) NOT NULL,
84 CONSTRAINT PatientLoc_PK PRIMARY KEY (pFileNumber , location),
85 CONSTRAINT PatientLoc_FK1 FOREIGN KEY (pFileNumber) REFERENCES Patient(fileNumber)
86 );
87
88 CREATE TABLE PatientPhone(
89 pFileNumber INT (6) NOT NULL,
90 phoneNumber INT (10) NOT NULL,
91 CONSTRAINT PatientPhone_PK PRIMARY KEY (pFileNumber ,phoneNumber),
92 CONSTRAINT PatientPhone_FK1 FOREIGN KEY (pFileNumber) REFERENCES Patient(fileNumber)
93 );

```

Create Tables

- takes
- visit
- visitor
- Views
- Stored Procedures
- Functions
- sakila
- sys
- unodb
- world

Administration Schemas

Information

No object selected

```

94
95
96
97
98
99
100
101
102
103
104
105
---
```

```

INSERT INTO hospital.MedicalStaff
VALUE (1120917163,'Assayil','Asaad','Hawsawi','female','dentalconsultant', null);
INSERT INTO hospital.MedicalStaff
VALUE (1007180207,'Ahmad','Massad','Alharthi','male','ophthalmologist',1120917163 );
INSERT INTO hospital.MedicalStaff
VALUE (1005695598,'Mohammed','Yahia','Albishri','male','internaldirector',1120917163 );
INSERT INTO hospital.MedicalStaff
VALUE (1120344163,'Anas','Jamal','Saggaf','male','surgydoctor',1120917163 );
INSERT INTO hospital.MedicalStaff
VALUE (1005695222,'Asmaa','Walid','Alats','female','pediatrician',1120917163 );
---
```

Use command insert Into to add information in all tables

| idNumber   | lName    | lName  | lName    | gender | specialization   | super_id   |
|------------|----------|--------|----------|--------|------------------|------------|
| 1005695222 | Asmaa    | Walid  | Alats    | female | pediatrician     | 1120917163 |
| 1005695598 | Mohammed | Yahia  | Albishri | male   | internaldirector | 1120917163 |
| 1007180207 | Ahmad    | Massad | Alharthi | male   | ophthalmologist  | 1120917163 |
| 1120344163 | Anas     | Jamal  | Saggaf   | male   | surgydoctor      | 1120917163 |
| 1120917163 | Assayil  | Asaad  | Hawsawi  | female | dentalconsultant |            |

medicalstaff 1 x

Output

Apply Revert

- sakila
- sys
- unodb
- world

Administration Schemas

Information

No object selected

```

106
107
108
109
110
111
112
113
114
115
```

```

INSERT INTO hospital.Patient
VALUE (1007180277,'Ibrahim','Abdullah','Alzahran','1991-12-5','male',112300 );
INSERT INTO hospital.Patient
VALUE (1120990361,'Najwan','Suleiman','Orif','2004-03-04','female', 115399);
INSERT INTO hospital.Patient
VALUE (1005698872,'Ghozlan','Akram','Busha','2000-11-8','female',112367 );
INSERT INTO hospital.Patient
VALUE (1120919876,'Sami','Asaad','Saggaf','1995-5-23','male',120100 );
INSERT INTO hospital.Patient
VALUE (1115695598,'Zahraa','Ali','Alshahri','1985-3-7','female',100364 );
---
```

| idNumber   | lName   | lName    | lName    | gender | birthDate  | lName  |
|------------|---------|----------|----------|--------|------------|--------|
| 1115695598 | Zahraa  | Ali      | Alshahri | female | 1985-03-07 | 100364 |
| 1007180277 | Ibrahim | Abdullah | Alzahran | male   | 1991-12-05 | 112300 |
| 1005698872 | Ghozlan | Akram    | Busha    | female | 2000-11-08 | 112367 |
| 1120990361 | Najwan  | Suleiman | Orif     | female | 2004-03-04 | 115399 |
| 1120919876 | Sami    | Asaad    | Saggaf   | male   | 1995-05-23 | 120100 |

patient 1 x

Output

Apply Revert

- Functions
- sakila
- sys
- unodb
- world

Administration Schemas

Information

No object selected

| idNumber   | lName   | lName    | lName    | gender | birthDate  | lName  |
|------------|---------|----------|----------|--------|------------|--------|
| 1115695598 | Zahraa  | Ali      | Alshahri | female | 1985-03-07 | 100364 |
| 1007180277 | Ibrahim | Abdullah | Alzahran | male   | 1991-12-05 | 112300 |
| 1005698872 | Ghozlan | Akram    | Busha    | female | 2000-11-08 | 112367 |
| 1120990361 | Najwan  | Suleiman | Orif     | female | 2004-03-04 | 115399 |
| 1120919876 | Sami    | Asaad    | Saggaf   | male   | 1995-05-23 | 120100 |

patient 1 x

Output

Apply Revert

- sakila
- sys
- unodb
- world

Administration Schemas

Information

No object selected

```

116
117
118
119
120
121
122
123
124
125
126
```

```

INSERT INTO hospital.Reception
VALUE (1197180203,'Ali','Saud','Theban','male');
INSERT INTO hospital.Reception
VALUE (1067760207,'Reem','Mohammed','Alharbi','female');
INSERT INTO hospital.Reception
VALUE (1143656457,'Amal','Yousef','Alharthi','female');
INSERT INTO hospital.Reception
VALUE (1144859027,'Lama','Abdulaziz','Alotaibi','female');
INSERT INTO hospital.Reception
VALUE (1159875650,'Batal','Omar','Alqhtani','male');
---
```

| employeeid | lName | lName     | lName    | gender |
|------------|-------|-----------|----------|--------|
| 1067760207 | Reem  | Mohammed  | Alharbi  | female |
| 1143656457 | Amal  | Yousef    | Alharthi | female |
| 1144859027 | Lama  | Abdulaziz | Alotaibi | female |
| 1159875650 | Batal | Omar      | Alqhtani | male   |
| 1197180203 | Ali   | Saud      | Theban   | male   |

reception 1 x

Output

Apply Revert

- sys
- unodb
- world

Administration Schemas

Information

No object selected

| employeeid | lName | lName     | lName    | gender |
|------------|-------|-----------|----------|--------|
| 1067760207 | Reem  | Mohammed  | Alharbi  | female |
| 1143656457 | Amal  | Yousef    | Alharthi | female |
| 1144859027 | Lama  | Abdulaziz | Alotaibi | female |
| 1159875650 | Batal | Omar      | Alqhtani | male   |
| 1197180203 | Ali   | Saud      | Theban   | male   |

reception 1 x

Output

Apply Revert

- sakila
- sys
- unodb
- world

Administration Schemas

Information

No object selected

```

128
129
130
131
132
133
134
135
136
137
```

```

INSERT INTO hospital.medicine
VALUE (3330544,'Adolisinus',25);
INSERT INTO hospital.medicine
VALUE (4320067,'After-meals',59);
INSERT INTO hospital.medicine
VALUE (507465,'Buscopan',30);
INSERT INTO hospital.medicine
VALUE (1276500,'Cidamex',100);
INSERT INTO hospital.medicine
VALUE (5566321,'Beptra',79);
---
```

| idNumber | medicineNumber | price |
|----------|----------------|-------|
| 507465   | Buscopan       | 30    |
| 1276500  | Cidamex        | 100   |
| 3330544  | Adolisinus     | 25    |
| 4320067  | After-meals    | 59    |
| 5566321  | Beptra         | 79    |

medicine 1 x

Output

Apply Revert

- world

Administration Schemas

Information

No object selected

| idNumber | medicineNumber | price |
|----------|----------------|-------|
| 507465   | Buscopan       | 30    |
| 1276500  | Cidamex        | 100   |
| 3330544  | Adolisinus     | 25    |
| 4320067  | After-meals    | 59    |
| 5566321  | Beptra         | 79    |

medicine 1 x

Output

Apply Revert

- sakila
- sys
- unodb
- world

Administration Schemas

Information

No object selected

```

139
140
141
142
143
144
145
146
147
148
```

```

INSERT INTO hospital.Visitor
VALUE (112300,'Maryam','Alharbi','female',25,'Wife');
INSERT INTO hospital.Visitor
VALUE (115399,'Suleiman','Orif','male',47,'Ffather');
INSERT INTO hospital.Visitor
VALUE (112367,'Samara','Busha','female',23,'Sister');
INSERT INTO hospital.Visitor
VALUE (120100,'Saud','Saggaf','male',28,'Brother');
INSERT INTO hospital.Visitor
VALUE (100364,'Rawan','Alharbi','female',44,'Mother');
---
```

| pFileNumber | lName    | lName   | gender | age | relationship |
|-------------|----------|---------|--------|-----|--------------|
| 100364      | Rawan    | Alharbi | female | 44  | Mother       |
| 112300      | Maryam   | Alharbi | female | 25  | Wife         |
| 112367      | Samara   | Busha   | female | 23  | Sister       |
| 115399      | Suleiman | Orif    | male   | 47  | Ffather      |
| 120100      | Saud     | Saggaf  | male   | 28  | Brother      |

visitor 1 x

Output

Apply Revert

- unodb
- world

Administration Schemas

Information

No object selected

| pFileNumber | lName    | lName   | gender | age | relationship |
|-------------|----------|---------|--------|-----|--------------|
| 100364      | Rawan    | Alharbi | female | 44  | Mother       |
| 112300      | Maryam   | Alharbi | female | 25  | Wife         |
| 112367      | Samara   | Busha   | female | 23  | Sister       |
| 115399      | Suleiman | Orif    | male   | 47  | Ffather      |
| 120100      | Saud     | Saggaf  | male   | 28  | Brother      |

visitor 1 x

Output

Apply Revert

patientphone

reception

takes

visit

visitor

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

149

150 • INSERT INTO hospital.takes

151 VALUE (112300,3330544);

152 • INSERT INTO hospital.takes

153 VALUE (115399,5566321);

154 • INSERT INTO hospital.takes

155 VALUE (112367,4320067);

156 • INSERT INTO hospital.takes

157 VALUE (112367,1276500);

158 • INSERT INTO hospital.takes

159 VALUE (100364,507465);

160

161

sys

unodb

world

AdministrationSchemas

Information

No object selected

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Contents

Result Grid

Form Editor

| pFileNumber | MedicineID |
|-------------|------------|
| 100364      | 507465     |
| 112367      | 1276500    |
| 112300      | 3330544    |
| 112367      | 4320067    |
| 115399      | 5566321    |
| 120100      | 507465     |

takes 1 x

Output

Apply

Revert

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

No object selected

160

161 • INSERT INTO hospital.Manages

162 VALUE (1120917163,3330544);

163 • INSERT INTO hospital.Manages

164 VALUE (1120917163,4320067);

165 • INSERT INTO hospital.Manages

166 VALUE (1120917163,507465);

167 • INSERT INTO hospital.Manages

168 VALUE (1005695598,1276500);

169 • INSERT INTO hospital.Manages

170 VALUE (1005695598,5566321);

171

sys

unodb

world

AdministrationSchemas

Information

No object selected

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Contents

Result Grid

Form Editor

| medicalID  | MedicineID |
|------------|------------|
| 1120917163 | 507465     |
| 1005695598 | 1276500    |
| 1120917163 | 3330544    |
| 1120917163 | 4320067    |
| 1005695598 | 5566321    |
| 120100     | 507465     |

manages 1 x

Output

Apply

Revert

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

No object selected

179

180 • INSERT INTO hospital.visit

181 VALUE (112300,1197180203,1007180207,500);

182 • INSERT INTO hospital.visit

183 VALUE (115399,1067760207,1120917163,200);

184 • INSERT INTO hospital.visit

185 VALUE (112367,1143656457,1005695598,750);

186 • INSERT INTO hospital.visit

187 VALUE (120100,1144859027,1120917163,150);

188 • INSERT INTO hospital.visit

189 VALUE (100364,1159875650,1005695598,300);

190

sys

unodb

world

AdministrationSchemas

Information

No object selected

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Contents

Result Grid

Form Editor

| pFileNumber | employeeid | medicalID  | price |
|-------------|------------|------------|-------|
| 100364      | 1159875650 | 1005695598 | 300   |
| 112300      | 1197180203 | 1007180207 | 500   |
| 112367      | 1143656457 | 1005695598 | 750   |
| 115399      | 1067760207 | 1120917163 | 200   |
| 120100      | 1144859027 | 1120917163 | 150   |
| 120100      | 1144859027 | 1120917163 | 150   |

visit 1 x

Output

Apply

Revert

manages

medicalstaff

medicine

patient

patientloc

patientphone

reception

takes

visit

visitor

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

No object selected

182

183 • INSERT INTO hospital.patientloc

184 VALUE (112300,'Alaziziyah ');

185 • INSERT INTO hospital.patientloc

186 VALUE (112300,'Alawali ');

187 • INSERT INTO hospital.patientloc

188 VALUE (115399,'Alhawqiyah ');

189 • INSERT INTO hospital.patientloc

190 VALUE (112367,'Alrashidiya ');

191 • INSERT INTO hospital.patientloc

192 VALUE (120100,'Alzahir ');

193 • INSERT INTO hospital.patientloc

194 VALUE (100364,'Altansim ');

195

reception

takes

visit

visitor

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

No object selected

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Contents

Result Grid

Form Editor

| pFileNumber | location    |
|-------------|-------------|
| 100364      | Altansim    |
| 112300      | Alawali     |
| 112300      | Alaziziyah  |
| 112367      | Alrashidiya |
| 115399      | Alhawqiyah  |
| 120100      | Alzahir     |

patientloc 1 x

Output

Apply

Revert

patientphone

reception

takes

visit

visitor

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

No object selected

196

197 • INSERT INTO hospital.patientphone

198 VALUE (112300,0552888163);

199 • INSERT INTO hospital.patientphone

200 VALUE (115399,0507343755);

201 • INSERT INTO hospital.patientphone

202 VALUE (112367,0566972247);

203 • INSERT INTO hospital.patientphone

204 VALUE (120100,0502772995);

205 • INSERT INTO hospital.patientphone

206 VALUE (100364,0558991226);

207 • INSERT INTO hospital.patientphone

208 VALUE (100364,0555854263);

visit

visitor

Views

Stored Procedures

Functions

sakila

sys

unodb

world

AdministrationSchemas

Information

No object selected

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Contents

Result Grid

Form Editor

Field Types

| pFileNumber | phoneNumber |
|-------------|-------------|
| 100364      | 055054263   |
| 100364      | 558991226   |
| 112300      | 552888163   |
| 112367      | 566972247   |
| 115399      | 507343755   |
| 120100      | 502772995   |

Output

Apply

Revert



245 \* **SELECT** \*  
 246 **FROM** hospital.Reception  
 247 **ORDER BY** fName **ASC**;  
 248

Use the Select command to display all columns from reception order by fName ascending

| employeeId | fName | mInit     | lName    | gender |
|------------|-------|-----------|----------|--------|
| 1197190203 | Ali   | Saud      | Theban   | male   |
| 1143656457 | Amal  | Yousef    | Alharthi | female |
| 1159875650 | Batal | Omar      | Alqatari | male   |
| 1144859027 | Lama  | Abdulaziz | Alotabi  | female |
| 1067760207 | Reem  | Mohammed  | Alharbi  | female |

249 \* **SELECT** \*  
 250 **FROM** hospital.Medicine  
 251 **ORDER BY** idNumber **DESC**;  
 252

Use the Select command to display all columns from medicine order by idNumber descending

| idNumber | medicineNumber | price |
|----------|----------------|-------|
| 5566321  | Bepira         | 158   |
| 4320067  | After-meals    | 118   |
| 3330544  | Adolatus       | 50    |
| 1276500  | Cidemes        | 200   |
| 507465   | Buscopan       | 60    |

253 \* **SELECT** CONCAT(v.fName, " ", v.lName) AS VisitorName, relationship, CONCAT(p.fName, " ", p.lName) AS patientName,  
 254 fileNumber, p.gender, birthDate  
 255 **FROM** Patient p **INNER JOIN** Visitor v **ON** p.fileNumber = v.pFileNumber  
 256 **WHERE** v.age >= 25;  
 257

Use the Select Command to display and contact the first and last name of patient ,and visitor name and their relationship, file number , gender and birth date and join the two table togher where visitors age greater or equal than 25

| VisitorName    | relationship | patientName        | fileNumber | gender | birthDate  |
|----------------|--------------|--------------------|------------|--------|------------|
| Razan Alharbi  | Mother       | Zahraa Alshahri    | 100364     | female | 1985-03-07 |
| Maryam Alharbi | Wife         | Ibrahim Alalzhrani | 112300     | male   | 1991-12-05 |
| Suleman Orif   | Father       | Najwan Orif        | 115399     | female | 2004-03-04 |
| Saad Saggaf    | Brother      | Sami Saggaf        | 120100     | male   | 1995-05-23 |

258 \* **SELECT** CONCAT(p.fName, " ", p.lName) AS patientName,  
 259 fileNumber, location , phoneNumber  
 260 **FROM** Patient p , patientloc pl , patientphone pp  
 261 **WHERE** p.fileNumber = pl.pFileNumber **AND** p.fileNumber = pp.pFileNumber;  
 262

Use the Select Command to contact the first and last name of patient with his file number , phone number and his location in one table using where condition to join the three table together

| patientName        | fileNumber | location    | phoneNumber |
|--------------------|------------|-------------|-------------|
| Zahraa Alshahri    | 100364     | Altanam     | 555054263   |
| Zahraa Alshahri    | 100364     | Altanam     | 558991226   |
| Ibrahim Alalzhrani | 112300     | Alawak      | 552888163   |
| Ibrahim Alalzhrani | 112300     | Alazayah    | 552888163   |
| Ghazlan Busha      | 112367     | Alrashidiya | 566972247   |
| Najwan Orif        | 115399     | Alhawayya   | 507343755   |

263 \* **SELECT** COUNT(phoneNumber) AS countOfphonNumber, pFileNumber  
 264 **FROM** hospital.patientphone  
 265 **GROUP BY** pFileNumber;  
 266

Use the Select Command to display the count of Patient phone number and their file number depending on file number

| countOfphonNumber | pFileNumber |
|-------------------|-------------|
| 2                 | 100364      |
| 1                 | 112300      |
| 1                 | 112367      |
| 1                 | 115399      |

267 \* **SELECT** \*  
 268 **FROM** Patient  
 269 **WHERE** fileNumber **IN** ( **SELECT** pfileNumber  
 270 **FROM** Visit  
 271 **WHERE** medicalID **IN** ( **SELECT** idNumber  
 272 **FROM** hospital.medicalstaff  
 273 **WHERE** fName = 'Mohammed' ) );  
 274  
 275  
 276

Use the SELECT command to display all columns for the Patient Who Visited Medical Staff named Muhammad

| idNumber   | fName   | mInit | lName    | birthDate  | gender | fileNumber |
|------------|---------|-------|----------|------------|--------|------------|
| 1115695598 | Zahraa  | Ali   | Alshahri | 1985-03-07 | female | 100364     |
| 1005698872 | Ghazlan | Akram | Busha    | 2000-11-08 | female | 112367     |